



Graft copolymerization of bio-based β -farnesene and *n*-butyl acrylate onto modified anionic cassava starch *via* emulsifier-free emulsion polymerization for sustainable adhesives

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ABSTRACT

In this study, bio-based β -farnesene (Far) and modified anionic cassava starch (ACS) were chosen as renewable alternatives to replace traditional non-renewable monomers in emulsifier-free emulsion polymerization for the development of bio-based materials. Graft copolymerization of Far and *n*-butyl acrylate (BA) onto ACS was carried out in a semi-batch emulsion polymerization, using 7.0 wt% of ACS and varying Far/BA ratios (100:0, 80:20, 70:30, 60:40, 50:50 and 0:100 wt%). By adjusting the Far/BA ratio, the final properties of the polymer latexes were effectively modulated. The use of ACS modified *in situ* proved colloidal stability of the latexes, with hydrodynamic diameter ranging from 150 to 200 nm and overall monomer conversion of 90–99 % for formulations containing up to 50 wt% Far. The gel content of poly(Far-co-BA)-g-ACS was significantly higher, while elongation at break decreased with increasing Far content. The glass transition temperature (T_g) of poly(Far-co-BA)-g-ACS ranged from -34 to $+3.2$ °C, which makes them attractive for a wide range of adhesive applications. The final polymers exhibited distinct adhesive performances, providing a strategic approach to tailoring end-use properties for specific applications. Furthermore, the use of *in situ* modified anionic cassava starch (ACS) provides valuable insights for the development of bio-based polymer materials under industrially relevant conditions.

1. Introduction

The development of bio-based polymer dispersions from renewable sources has gained significant attention in recent years due to growing environmental concerns to replace nonrenewable petroleum-based monomers, such as 1,3-butadiene and isoprene, avoiding the use of expensive pressurized reactors for a broad range of industrial applications, including paper coatings, elastomers, adhesives, and others (Behr & Johnen, 2009; Driesbeke et al., 2020; Luk & Maric, 2019, 2021; Rigo et al., 2023; Rodrigues & Oliveira, 2024; Sahu et al., 2023; Sahu & Oh, 2022). The global production of bio-based polymer latexes and the use of bio-based materials in the latex formulation reflect in the reduction of the synthetic polymer content where possible and further increase sustainability (Mohebbi et al., 2025).

The use of modified cassava starch in water-based polymer dispersions as an alternative to traditional surfactants and nonrenewable monomers has gained more attention due to its low cost, biodegradability, availability, and renewability (Cazotti et al., 2020; Rigo et al., 2023; Rodrigues & Oliveira, 2024). Starch, one of the most abundant

polysaccharides present in plants, consists primarily of two major components: amylose and amylopectin. Amylose is composed of linear chains of glucose units linked by (α)-1,4-glycosidic bonds, while amylopectin consists of branched chains of glucose units linked by α -1,4-glycosidic bonds (Rashwan et al., 2024; Zeeman et al., 2015). Starch can be sourced on a large scale from various plants, including rice, sorghum, potatoes, cassava, and corn (Zeeman et al., 2015). Starch from potatoes, corn, and cassava typically has an amylose content ranging from 20 to 30 % in regular varieties, with high-amylose varieties containing up to 80 %. Cassava roots are a crop in many tropical regions and typically contain high starch content (60 % to 80 % by dry weight), predominantly composed of amylopectin (Rashwan et al., 2024; Wang et al., 2022). Therefore, modified starch is a suitable candidate for an eco-friendly replacement of petroleum-based polymer dispersions.

Graft copolymerization of starch is a crucial approach to enhance the physicochemical properties of starch and can be achieved *via* grafting from, grafting to, or grafting through approaches (Cazotti et al., 2020; Meimoun et al., 2018; Rodrigues et al., 2021; Smeets et al., 2017). Furthermore, various drawbacks were still present, such as latex

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